PROPOSAL:
GRENADA TECHNOLOGY & INNOVATION HUB

Today's business leaders demand a workforce of highly skilled IT specialists to design, manage, and maintain their information systems. As the convergence of industry and technology matures, open innovation will provide a reservoir of well-paid IT professionals who are prepared to step in and are ready to deploy resolutions to complex technological challenges at a moment's notice.

Through the development of a dedicated Technology and Innovation Hub, Grenada would develop both a highly skilled workforce, and the business leaders looking to employ them.

CONTENTS

ACKNOWLEDGMENTS

This document was prepared with contributions from Clifford Browne , Giselle Frederick, , Sascha Williams, Filip Jakubowski, Brendan Heyck, Kitaka Mawuto, Adrian Sayers, Dr. Carl S. Roberts, George Jr Matthew, and Richi Ramlal, collectively referred to throughout this document as 'the team' or 'the exploratory team'. Further information about each contributing member is found at the end of this document.

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Please note: The views represented in this publication are those of the author(s) and do not necessarily represent those of the United Nations, including UNDP, or the UN Member States.

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EXECUTIVE SUMMARY

In the wake of global pandemic and other shocks, Grenada has felt the challenges of relying too heavily on the tourism sector. A Technology & Innovation Hub in Grenada would deliver hope for the country, new opportunities for local youths, keeping the most highly trained and intelligent people in Grenada. It can promote innovation and a creative environment, and enable a generation to thrive based on knowledge, skills, open competition, specialization, increased capacity, and efficiency.

A Technology and Innovation Hub is a place where collective understanding of opportunities and challenges are solved using emerging technologies and innovations; a place where resources are shared and strengthened, where new products (product innovations) or processes in application (process innovations) have been developed or created and brought to market. Key components of a Technology and Innovation Hub are sources of funding, access to venture capitalists, angel investors, and government sponsorship.

This study builds upon past research, case studies and extensive consultations to offer a vision and a roadmap for a Technology & Innovation Hub in Grenada. It proposes a Technology and Innovation Hub for Grenada that will:

- Establish and cultivate a culture in which young people and those new to running technology businesses will be supported to establish their own start-ups
- Promote a strong and supportive business environment which directly and indirectly will provide full-time jobs to the public, private, education, and research and development sectors.
- Provide access to capital so that entrepreneurs may build enterprises and businesses throughout the region, as well as virtual and digital workspaces in which to undertake the entrepreneurial and collaborative efforts.
- Provide recommendations for modernisation of Grenada's primary, secondary and tertiary schools' curriculum by inclusion of courses like coding, robotics, data science, tech in tourism and agriculture, and cyber security
- Provide programs and services to help small and medium companies create innovative products and bring those products to market.
- Provide evidence-based recommendations for policy changes such as in terms of taxation, logistics, visas and immigration, education, and cooperation with other sectors such as banking to suggest financial instruments that can support this undertaking and encourage commitments from the private and public sectors, as well as from the diaspora

As the proposed approach for the institutional arrangement for the Technology and Innovation Hub, the team recommends the creation of a statutory body, established by an act of parliament and authorized to implement certain services and projects on the behalf of the Government of Grenada. In the interim, the creation of a non-profit entity for the Technology and Innovation Hub, could serve to advance this agenda in partnership with the Government of Grenada with an administrative desk or taskforce within the ICT department assigned to work closely with the Technology and Innovation Hub. This approach for the legal structure would guarantee longevity and the success of Grenada Technology and Innovation Hub for the benefit of all Grenadians.

BACKGROUND

Within its 2035 National Sustainable Development Plan Grenada recognizes its small size as a distinct advantage. At the same time, Grenada's main economic drivers are positioned in and around its physically vulnerable coastal capital city on the frontlines of climate change. In recent years, there has been increased interest in building Grenada's economy in a sustainable and climate-resilient manner. This will support the country to develop in an ecologically, economically and socially sustainable manner. Whilst having a strong vision that attracts investment from domestic and international sources, rooted in achieving an ambitious sustainability agenda that is green, resilient and globally relevant to a technologically advancing era.

In this context, in late 2020-21 UNDP Barbados and the Eastern Caribbean Multi-Country Office supported the Government of Grenada to conduct a rapid assessment of the country's readiness for digital transformation to facilitate developmental leapfrogging and resilience building. The report aimed to define a vision of a Smart Small State for Grenada, including defining the government's priority areas for digitalization, identifying comparative approaches and good practices around emerging technologies for all relevant sectors of Grenada's' economy. Among the report recommendations were the following:

- Grenada should leverage its population size and territorial extension and think of itself as a smart city, allowing for more cost efficient, agile, and nimble capacity to experiment with different smart city solutions and concepts.
- Grenada needs to work together with the private sector to drive this transition by having an alliance approach, where established technology leaders work in partnership with the Government to develop the Smart Small State concepts, allowing the government to provide incentives for SME's to digitize, develop climate smart solutions, and increase financial inclusion through the country.
- Grenada should tap into the Grenadian diaspora living abroad to contribute to Grenada's development, particularly in building the human capital of the country.
- **Grenada must establish an Innovation Hub to develop the tools, competences, and skill-sets of local businesses and people.**

This current study is a direct follow up to this work. Conceptual discussions for the creation of an ICT Hub have been underway in Grenada for sometime, however, a clearer analysis of options and lessons to drive this development was needed to explore, analyze and provide suitable and sustainable options to the Government of Grenada in planning for an adoption of a national business and ICT hub. This aims to complement the government's ongoing digital transformation efforts and may highlight the need for the national coordinated effort toward development of a national ICT hub. The scope of this research has drawn upon this Smart Small State study in Grenada, as well as the Accelerate Grenada initiative and the UNDP Fut-tourism Regional Dialogues report.

INTRODUCTION

The confluence of shifting perspectives on both macro and micro scales presents a unique opportunity for Grenada to emerge with a stronger, more diversified economy through the development of new programs and initiatives, in this case a dedicated Technology and Innovation Hub. At its core, a Technology & Innovation Hub in Grenada could deliver hope for the country, new opportunities for local youths, and should keep the most highly trained and intelligent people in Grenada. It can promote innovation and a creative environment, and enables a generation to thrive based on knowledge, skills, open competition, specialization, increased capacity, and efficiency.

There is perhaps no greater understatement than to say that COVID-19 has changed the world. Along with dramatic shifts in healthcare, work culture, travel trends, supply chains, and more, new realities are emerging. For example, on a macro level the world has seen a shift to distributed workforces and a desire to spread out, find space, and explore new opportunities. On a micro level, Grenada has felt the challenges of relying too heavily on one industry, in this case tourism.

Grenada and the Southern Caribbean are blessed with an attractive climate and welcoming topography, as well as some of the friendliest people in the world. It is no surprise that tourism thrives on the island. However, global turmoil, caused by events like COVID-19 and the financial crisis, as well as localised challenges such as hurricanes and their associated natural disasters, show how challenging it is to rely on a single outsourced industry.

This document aims to improve the understanding of the benefit of a Technology & Innovation Hub to all sectors of Grenada's economies. In addition, the team worked to analyse, design, and propose suitable options to the Government of Grenada to assist with their planning for the execution of a Technology & Innovation Hub in Grenada. Establishing a Technology & Innovation Hub complements the government's ongoing digital transformation efforts and highlights the importance of a coordinated effort to develop a national technology and innovation strategy.

Through this document, the team will explore and suggest how policy makers could make Grenada's Technology & Innovation Hub a reality and how this initiative could bring common prosperity for all Grenadians. The objectives of this Technology & Innovation Hub study are as follows:

- To provide an overview of the concept of a Technology & Innovation Hub and its proposed place in Grenada
- To examine past case studies (i.e.: Israel, Costa Rica, South Korea, Kelowna) and compare them to the particularities of Grenada's situation
- To analyse potential programs and partnerships both within Grenada and internationally
- To describe a roadmap towards completion, discussing both virtual and physical spaces and programs, to explore potential roadblocks, and to provide financial guideposts for the project

CASE EXAMPLES

History has shown that in Grenada, by using the power of collaboration among private and public sectors, citizens and residents, Grenadians in the diaspora, and friends of Grenada, citizens can take any concept or initiative from an idea to a massive income-generating project.*

A first step to initiate this research was to explore lessons from other countries and cities that have successfully shifted towards technology as a key economic driver. We looked at Israel, South Korea, Costa Rica, Kelowna and Scotland and found some key takeaways around past successes and initiatives.

It must be noted that both Israel and South Korea were able to transform their economies with limited natural resources; these countries invested in technology and human resources development with a focus on science, engineering, math and technology. Today, South Korea is ranked number 5 in the Economic Complexity Index and Israel is ranked number 20.

In the case of Israel, the three major drivers behind their high-tech community are strong partnership with universities, a vibrant start-up community, and access to capital.




The World Bank also funded a project in South Korea to boost the transition to the green and sustainable economy and scale-up data-driven smart agriculture. Costa Rica was also able to transform its economy through Tech Innovation. The three strategies for their innovation hubs were as follows:

- Physical resources (for example hardware and software (internet) availability) for the entire population
- Human resource development with regards to tech
- Cyber security workforce development

*For example- the Grenada Co-operative Bank which is a locally owned and operated bank on the island and maybe the biggest bank on the island was once Grenada Penny Bank. The bank was able to market and promote itself as a local owned bank and using the strategy of patriotism and targeting the private and public sectors as well as the diaspora to invest in a local owned and operated bank, today Grenada Co-operative Bank is both a success story and very profitable financial institution. The Bank continues to report another six months of good performance with a profit after tax (unaudited) for the six months ended March 31, 2021, of $6.2M. The reported profits were greater than forecasted but declined by 15% or $1.1M compared to the previous corresponding six months profits of $7.3M. Understandably, this performance has been affected by the impact of the COVID-19 pandemic on both the local economy and Bank's operations.

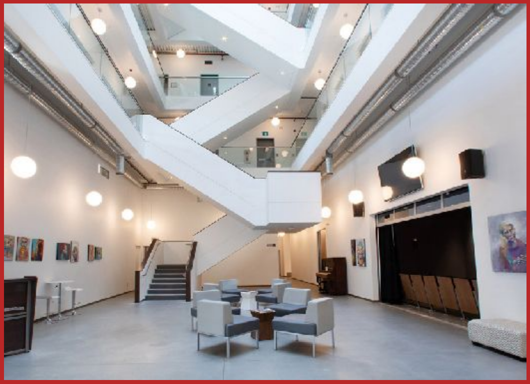
PHYSICAL SPACE CASE STUDY: KELOWNA INNOVATION CENTRE

The Kelowna Innovation Centre in British Columbia, Canada offers a successful example of what a thriving tech hub can look like. The Innovation Centre is the Okanagan's hub of technology, entrepreneurship and creativity and houses everything from solopreneurs (one-person start-ups) to large technology and innovation firms. Publicly-supported spaces and services are also available for early-stage companies, non-profits, community groups, and social enterprises.

The Kelowna Innovation Society (KIS) is a government-funded not-for-profit with a mandate to foster the economic diversification of the Central Okanagan's regional economy through the support and promotion of innovation, technology and entrepreneurship. KIS works closely with all levels of government and with entrepreneurs, industry, academia and the private sector to deliver on this mandate.

One of the key initiatives for KIS was to establish and develop space within the Innovation Centre specifically designed to support entrepreneurs and founders of early stage companies and supporting organizations from the community.

Design, functionality and environmental sustainability were all driving influences during the creative process that became the Innovation Centre. Influenced by the intentional designs of the Google, Apple and Pixar campuses, the Innovation Centre is purposefully built to encourage unplanned encounters and creative collisions. These are facilitated by several shared and community spaces: a floor to ceiling atrium, a rooftop patio and restaurant, a magnificent white staircase and a mezzanine that wraps each floor.



Information and photos: Kelowna Innovation Centre, https://kelownainnovationcentre.com

Costa Rica's strategy was based on innovation, research and development, productivity, human resource development, and tax exemption for all computer and hardware devices. Costa Rica was also able to implement a free trade policy that resulted in an increase in employment and revenue for the country.

Taking into account the above examples, as well as research into further arenas including the EU-CELAC 2021-2023 Strategic Roadmap for the implementation of the Brussels Declaration and EU-CELAC Action Plan on Science, Technology and Innovation, the team highlights four key ingredients:

- A single-minded dedication to recruiting key experts in the fields of science and engineering to be part of the Technology & Innovation Hub

- An emphasis on higher education, above all, excellence in math as a foundation for educating engineers and scientists

- Establishment of a tech infrastructure that focuses on smart tourism and smart farming,

- The adoption of an entrepreneurial focus to grow an innovation ecosystem through a foundation of training and mentorship.

BENEFITS OF IMPLEMENTING A TECHNOLOGY & INNOVATION HUB IN GRENADA

Technology and Innovation Hubs in Israel, Mexico and Costa Rica are responsible for the creation of thousands of high paying, highly skilled jobs and are contributing billions of dollars and increasing digitization of the economies of these countries annually, through greater connectivity, and the development of exportable skills.

Technology and innovation hubs across the world are shaping and changing work, and transforming labor markets. The Government of Grenada and other stakeholders can respond to leverage the growth of a technology and innovation hub to help increase employment opportunities.

This initiative is very timely for our tri-island state of Grenada, Carriacou and Petite Martinique; the hub can enable employment, will help address some of the very high unemployment and underemployment problems, both by creating jobs in the ICT sector, and by helping to make labor markets more inclusive, innovative, flexible and transparent. A technology and innovation hub is a strategy to enable employment opportunities in countries around the world that are looking to hire highly skilled tech workers. This will result in positive economic and social implications not only for the individual but for his/her family and by extension Grenada's economy.

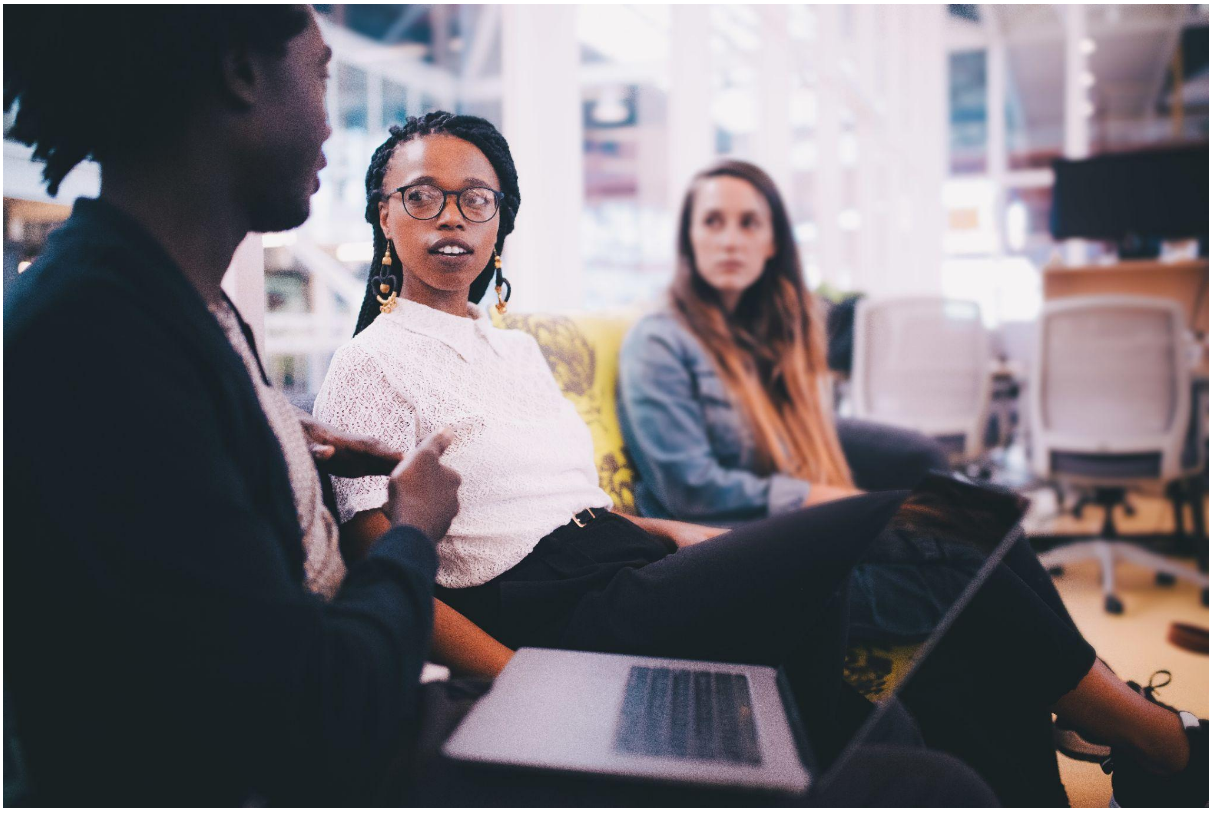
A technology and innovation hub can empower workers by making labor markets more transparent, innovative, and inclusive. This initiative can make it easier for workers in Grenada to find jobs locally and internationally and for employers to find skilled workers. The hub can augment digitized work, enable innovation, and has the potential to connect workers to work irrespective of their location; it is possible that tech hubs could help overcome the social, cultural, and physical barriers that might otherwise have excluded women or people with disabilities from participating in the labor market.

It is in Grenada's strategic interest to build a digital community culture and increase the number of persons being trained in Tech, as well as to develop and use technology to improve lifestyle and provide common prosperity for all.

With—as of 2021—112,000 citizens residing on the island of Grenada, 60% internet penetration, and 117% mobile phone penetration as a percentage of the population, there is no shortage of potential tech workers and entrepreneurs on the island, ready to take advantage of such a project.

Exploration into technological interest has already begun. In late 2020, the UNDP Barbados and the Eastern Caribbean Multi-Country Office supported the Government of Grenada to conduct a rapid assessment of the country's readiness to become a Smart Small State in all relevant sectors where digital transformation could facilitate developmental leapfrogging and resilience building, including a focus on exploring possibilities for blockchain technologies.

The study called for digital innovations to be piloted to build momentum for the Smart Small Grenada vision and demonstrate the potential for digital transformation in key sectors. Under the GEF/UNDP Climate Resilient Agriculture Project, a digital challenge was launched to support innovators in the agro-processing sector in efforts for digital traceability and blockchain and digital marketing. The central theme demands the requirement of an urban ecosystem containing the application of digital and information technologies to provide innovative and intelligent solutions such as infrastructure, governance, security, transportation, healthcare, e-business, e-tourism, and education.



DEFINING INDUSTRY 4.0

According to *Forbes*, "[w]e're in the midst of a significant transformation regarding the way we produce products thanks to the digitization of manufacturing. This transition is so compelling that it is being called Industry 4.0 to represent the fourth revolution that has occurred in manufacturing…" They go on to describe how Industry 4.0 refers to a period where "computers are connected and communicate with one another to ultimately make decisions without human involvement."

Industry 4.0 requires effective instruments to reshape the way in which a country's economic sectors work. A digital Technology & Innovation Hub would aid in the adoption and understanding of large and small companies in their digital transformation opportunities and benefits.

The 2016 World Economic Forum discussed the 4th industrial revolution relative to technological convergence and open innovation as leading the new dynamic economy and sustainable development (Park, 2017). Experts generally agree that the creation of new market demand through convergence, openness, and disruptive technologies such as Artificial Intelligence (AI), big data, and internet of Things (IoT) may contribute significantly to solving the current global economic crisis.

Generally, a trend or phenomenon where two or more independent technologies integrate and form a new outcome is referred to as technological convergence. A good example of a device that integrates several independent technologies is the smartphone—television, camera, music player, computer, navigation and geolocation tool—all in a single device. The smartphone is a $350 billion industry that has become its own identifiable category of technology.

To take advantage of the opportunities presented by the Technology & Innovation Hub and the burgeoning Industry 4.0 revolution, Grenada will need to embrace and enhance new technologies. Some such technologies include:

AGRITECH *A global Community of Practice, where people from all over the world exchange information, ideas, and resources related to the use of technology for sustainable agriculture and rural development.*

ARTIFICIAL INTELLIGENCE (AI). *In its simplest form, artificial intelligence is a field which combines computer science and robust datasets to enable problem-solving. It also encompasses sub-fields of machine learning and deep learning, which are frequently mentioned in conjunction with artificial intelligence.*

DIGITAL ENTREPRENEURSHIP. *Creating a business on the internet; selling services or products online, without the need to invest in physical spaces. A few examples of digital businesses are online courses, e-commerce, blogs, YouTube channels, and technological solutions in general.*

ELEARNING. *A learning system based on formalised teaching but with the help of electronic resources is known as E-learning. While teaching can be based in or out of the classrooms, the use of computers and the internet forms the major component of eLearning.*

ETOURISM. *The digitization of all the processes and value chains in the tourism, travel, hospitality and catering industries that enable organisations to maximise their efficiency and effectiveness.*

The promise of an information infrastructure and Technology & Innovation Hub comes with greater ease and access to an enormous amount of information and innovation. The potential to share information and knowledge may create positive social change by lowering barriers to new inventions, business and banking processes, establishing eCommerce, and increasing online e-learning opportunities.

The benefits to citizens include enhanced living conditions and self-empowerment; thus, improving their socio economic position. Empirical data from European Commission, article "Shaping Europe's Digital Future" indicates substantial advantages of a digital innovative hub to the citizens and society of a developing smart state.

The wide use of computer networks to facilitate tech, the ubiquity of information in a digital form, along with the extensive proliferation and high usage of the internet has intrinsic advantages to the process of conducting business, and how new paradigms of innovation are crystalised in the development of rural and urban sectors of society.

The Government of Grenada could enjoy a more highly-educated tax and consumer base, the reversal of so-called 'brain drain', and an expansion of Grenada's impact on the world stage.

Additionally, the Technology & Innovation Hub will incentivise foreign tech workers, digital nomads, business owners, and investors to move capital and information into the country.

For expatriate Grenadians, this program may provide the incentive and ability to reconnect in a more meaningful business and investment sense with their homeland. For young Grenadians, the improving and expanding tech sector will provide a pathway to remaining in-country while also enjoying stable, well-paying, and fulfilling careers.

At a glance, the Technology & Innovation Hub:

- Creates a geo-transitioning axis to a knowledge society
- Provides a centralised hub where IT project managers, business managers, IT professionals, and researchers may find solutions to complex technical problems
- Directly improves the lives of ordinary citizens
- Leverages a ripple effect on the economic growth of the country through increased business opportunities and a greater workforce.

PROPOSED PROGRAMS

The vision for the Grenada Technology & Innovation Hub is rich. From virtual spaces to a physical home base, the team envisions a holistic program which draws in foreign entrepreneurs, encourages Grenadians to stay in-country, and begins educating the youngest of tech entrepreneurs through enriching education programs at all levels of schooling.

As mentioned, a challenge currently facing Grenada is an over-dependence on tourism (and agriculture). Thus, this project aims to expand economic growth into the following arenas:

- **AGRITECHNOLOGY**—Agritechnology's skilled team uses science to develop new products and processes for clients in agriculture, nutraceuticals, food manufacturing, biofuels.
- **SUSTAINABLE TOURISM**—Sustainable tourism is a concept that covers the complete tourism experience, including concern for economic, social and environmental issues as well as attention to improving tourists' experiences and addressing the needs of host communities.
- **BLUE ECONOMY**—Blue economy is relating to the sustainable exploitation, preservation and regeneration of the marine environment).
- **RESEARCH, DEVELOPMENT AND INNOVATION**—The focus is technology that is impactful and sustainable for small island states and Small Medium Size Enterprise (SMEs).

This vision can be realised in as little as three years with initial benefits being achieved just six months from inception. **The team proposes implementing the Hub in three phases.**

STEP ONE — SHORT TERM

Step one would allow for high-impact, low-cost initiatives such as developing courses and building a tech talent pool. It would also be a period of relationship building.

STEP TWO — MEDIUM-TERM

In stage two, partnerships would be formalised and international partners would become more involved. Larger plans would be developed. A virtual Hub would be increasingly developed.

STEP THREE — PERMANENT

As the Hub grows, so to will the need for a dedicated space. In this stage, larger capital contributions will be required to build a physical building as well as continuing the work from past stages.

PHASE ONE

The first phase will establish partnerships and collaborations with educational institutions and small and medium-size enterprises (SMEs) to provide training workshops and awareness programs; for example tech mentorship, job training and hackathons. Partner organizations could include the Department of Education, UNDP, UNESCO, USAID, tourism authorities, schools and training organizations already in place, and further ministries, NGOs, and public—and private-sector bodies.

This phase will also see the completion of several initial logistical tasks, such as:

- Proposing details about the selection and preparation of the Centre's location
- As an interim step towards formalization, the potential incorporation of a non-for-profit legal entity
- Signing a Memorandum of Understanding (MOU) with all partners, stating resource commitments; to be announced.

Some early-stage milestones, such as forming a steering committee and appointing an exploratory team have already been completed.

PHASE TWO

The second phase will see the formation of commercial partnerships between businesses and companies mentioned in the partnership section to provide fintech opportunities, research and development, blockchain apps and wallets for businesses, crypto-currency, NFT (a non-interchangeable unit of data stored on a blockchain, a form of digital ledger, that can be sold and traded), etc. As an example partnership, Brazen Group Inc. is prepared to deliver training and workshops in these areas.

The **THIRD PHASE** will foster innovation by implementing more significant policy shifts, establishing pathways to prove and incubate innovation ideas and concepts. The third phase will also aim to introduce innovation-friendly policies, in partnership with government, to encourage inward investment and skilled talent attraction. It would be a logical phase in which to explore the building of a dedicated physical space and the transition towards the creation of statutory body for the Hub.

Through all phases, the Technology & Innovation Hub will be working towards an Open Innovation Strategy (OIS), which aims to redefine the boundaries between the company and its environment, making the Technology & Innovation Hub more porous and embedded in a loosely-coupled network of different actors. Collectively and individually they will be working towards creating and commercialising new knowledge in one open space.

Open innovation is about involving more actors in the process, from researchers to entrepreneurs, to students, to governments and civil society. This approach reduces the capital investment and integrates both the nation and the region into one digital space. Open innovation is needed to capitalise and maximise on the outcomes of Tech innovation. An Open Innovation Strategy allows stakeholders to share experience and knowledge with other countries that are pursuing a Technology & Innovation Hub Initiative. (For example, Dominica has expressed an interest in pursuing a project of this nature.)

The team believes that positioning the Technology & Innovation Hub in a way that allows **Open Innovation, Open to Research and Open to Employment and Workforce Development Opportunities** will ensure sustainable futures for Grenada's future generations.

PHYSICAL SPACE

While the early and immediate stages of the Technology & Innovation Hub concentrate on digital communities and spaces with education and outreach programs that may exist solely online, a key component of Grenada's Technology & Innovation Hub will eventually be a physical hub. This space is positioned as a future cornerstone of the region.

Utilising a self-sustaining and solar energised building, the team envisions a space capable of supporting the following facets of technological innovation: E-commerce & Marketplace Development, Business Intelligence & Analytics Solutions, Software Development, Digital Marketing Solutions, Web Application Development, Game Modelling & Simulation, Website Design & Development, Blockchain, AI and Robotics showcase, etc.

The Technology & Innovation Hub could include business offices to facilitate tech startup businesses and vendors, classroom spaces, boardrooms with the capacity to adapt to a mobile workforce, on-site accommodations, a restaurant, R&D SmartLab, and eco-friendly features such as a green roof and solar parking facility.

The Technology & Innovation Hub can be viewed as a business incubator for technological entrepreneurship, and one of its goals should be to help technology-based entrepreneurs successfully develop their products and enable them to transition into the financing stage and enter the market. It could also aim to support the country's universities in their research, learning, teaching and administrative processes.

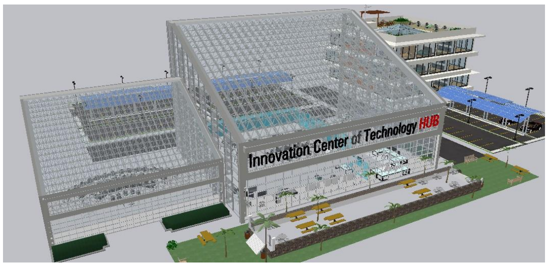
The space could also work as a nexus for the initiation, planning, and execution of programs aiming at increased quality and productivity of the local tech industry.

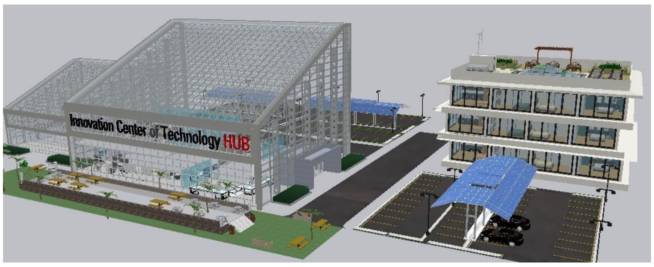
The Technology & Innovation Hub's approach to delivering training should be based on the tenets of adult learning styles including teamwork, project-based work, and Open Innovation strategies. Moreover, the hub could offer "blended training programs" that focus on the interoperability of different systems from different vendors. This hands-on training emphasises real-work scenarios, projects and research and development.

The Technology & Innovation Hub would also focus on skill and capacity building for workers within the public and private sectors and for students in colleges and universities. The duration of these training programs, workshops and research development courses could be as short as one to two days, while still delivering incredible value.

Longer training and innovation programs like hackathons, innovation challenges, research and development summer camps would also be delivered within the Tech ecosystem.

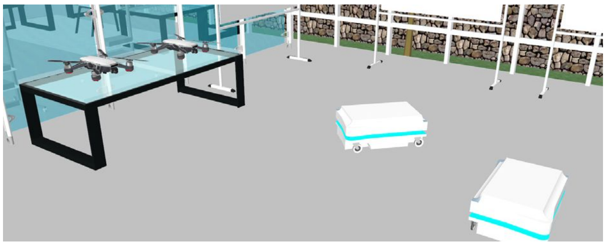
Following are possible renderings on what an innovation space could look like, with the vision of engaging innovation in new and emerging technologies within Grenada and the region at its forefront.





Engagement is important to drive innovation and problem solving through collaboration and partnership. This tech space will showcase competence development in all areas of industry 4.0, and will spark an innovative cultural mindset across the Caribbean. Promoting the use of renewable energy, the Technology & Innovation Hub may provide R&D and proof of concept projects to the various industries across the country and region.

Though the physical space is anticipated as part of a future phase, below the team has described some potential impact areas which could be further developed by the hub.

AUTOMATION & ROBOTICS

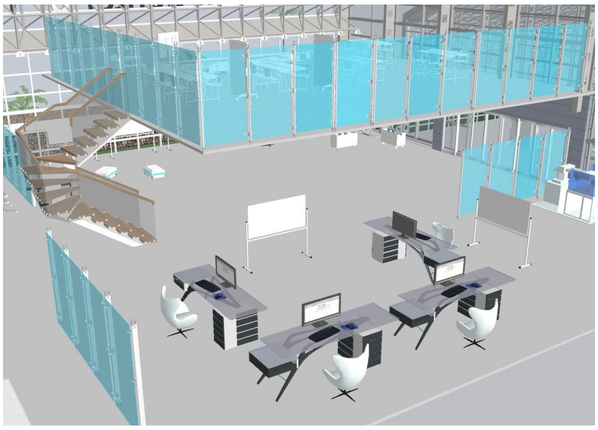
The Technology & Innovation Hub automation and robotics domain may explore and manage projects that will advance remedies to specialised labour shortages and safety issues. Policy makers in the agricultural sector, as well as agronomists, would obtain a better understanding of the advantages and constraints of robotic farming.

The world is moving towards electromobility and autonomous vehicles. The Technology & Innovation Hub may provide proof of concept and advancements to planning, while making improvements to the IT infrastructure. In turn, this enables the adaptation and acceptance of such technologies.

VIRTUAL/AUGMENTED REALITY & SIMULATIONS

What will Grenada look like in 2025?

This building will provide space for the simulation of “real life” scenarios and high-tech innovation. Technology simulation competence could allow persons to see prototypes of products, machines, services, and transportation before it becomes an implemented reality.



ARTIFICIAL INTELLIGENCE (AI) & MACHINE LEARNING

The Technology & Innovation Hub may provide data-centric solutions to healthcare, education, fire dynamics, law enforcement, and much more. From traffic pattern predictions to police allocations, AI has the ability to deliver unprecedented, predictable innovations, leading to vast improvements for society.

No matter the specific spaces and areas of development, the Hub would be guided by the following principles: encompassing rapidly emerging, evolving, and converging computer, software, networking, telecommunications, internet, programming, information systems, and digital media technologies.

Additionally, the use of Enterprise-grade fibre optic infrastructure/backbone, equipment, hardware, software, and networking solutions, will enable reliable access to the best qualities of tech training and e-services demands. **Emphasis on procedures, standards, policies, and security will facilitate a safe, successful, and satisfying experience for all users.**

PROPOSED PARTNERS

A variety of stakeholders must be involved in all stages of the Technology & Innovation Hub development program for it to be a success. Momentum is growing for the concept of the Hub in Grenada and discussions with stakeholders have revealed several potential leads of interest. For instance, initially these could include the organisations listed in the Phase One description in Section Two. With perseverance, hard work and mentoring, and with the right regional and international partners—especially from the diaspora—Grenada's Technology & Innovation Hub would succeed. The Hub would partner with colleges and universities, tech start-ups, tech academies, international development partners, and industries locally and throughout the region.

The Caribbean Coding Academy, as a proposed founding partner in the Hub, would offer awareness and training in the areas of coding, robotics and robotic process automation. They could assist the Department of Technology in receiving Cyber Security training with one of their international partners: **SecureXperts**. This cyber security company is also available to provide workforce development training to upper level high school and college students.

The Technology & Innovation Hub will co-host activities such as research, hackathons and training workshops in collaboration with UNDP and other international development partners that can offer both financial and technical support to such efforts.

Potential regional and international companies are already identified as fee-based training partners for this initiative. For instance, there have been expressions of interest from cyber security company **SecureXperts** Inc and US-based **Black Pearl** to provide cyber security training in both their hardware and software products and services. Black Pearl is currently providing cyber security services and products to the United States Department of Defence and NASA.

UiPath, a Robotic Process Automation (**RPA**) firm and **Green Light Consulting** from Canada have persons receiving training in the areas of robotics and artificial intelligence on the island. These companies are currently working on the ground in Grenada, providing tech training and mentorship.

Throughout the course of this research the team has also identified potential private international donors and successful Grenadians residing abroad who are interested in contributing financially to this project. These donors have the resources and the human capital to help in the sustainability of the Technology & Innovation Hub; their roles will be to establish R&D centres for the purpose of knowledge transfer and innovation within the hub ecosystem.

In addition to international donors, local and regional private and public companies can provide financial contributions based on "Fee for For Services." These services can be in technology training and research and development.

TECHNOLOGY TRAINING PROVIDERS: THE CURRENT LANDSCAPE

Overall there is a lack of educators and training providers in the tech space on the island. However, with support and development, each of the following training providers could become important partners for the Hub. A successful innovation hub will also partner with junior colleges in St. Vincent, St. Lucia, and Dominica to name a few, as well as universities throughout the region and internationally to conduct research and innovation on the island. The objective is to get the brightest and most skilled youths in the Caribbean to be part of this Technology & Innovation Hub. Through engaging with the educational entities below, the Hub can work with existing stakeholders in the region to improve the breadth and uptake of technology-based education. These partnerships might include sharing spaces, resources, and teachers and experts, as well as providing a pathway for students interested in careers and entrepreneurship within the realm of technology.

Primary and Secondary Schools

While Information Technology is taught in secondary schools, the contents or syllabi aren't sufficient and detailed enough to provide a strong foundation that can influence students' career path and professions. That said, it is the Ministry of Education's short term goal to include coding and robotics as part of the primary and secondary school curriculum. Introductory courses in CRM software, SaaS platforms, Web3 Technologies, artificial intelligence, etc would also help to develop young tech entrepreneurs.

TAMCC

T. A. Marryshow Community College is one of the leading providers of Higher Education and Supplementary Education in Grenada. The junior college offers Associate Degrees in Information and Technology and other Information Technology related programs. Students graduating from the college usually have a strong foundation in coding, the majority of these students continue their studies at St. George's University. The college has a yearly Information and Technology graduation class of 35 students annually.

St. George's University

St George's University is the island's premiere university. Established in 1978 on the island of Grenada, the primary objective of the University is to educate and train students in the field of medicine and veterinary science. The secondary objective of the school is to educate and train students in the field Arts and Sciences, which includes Information and Technology. The school of Art and Science started in 1996 and since then the school has been offering a Bachelor of Science Degree in Information and Technology. The university has an average of 28 students graduating annually with a Bachelor of Science Degree in Information Technology.

Caribbean Coding Academy Grenada

Founded in 2009 under a different company name (Ameltek) and rebranded in 2018 as Caribbean Coding Academy. The academy is a private organisation providing non-accredited training of technical skills in Grenada in the fields of software development, Cyber Security, Robotic Process Automation, Software as a Service, Salesforce and other tech-related job training programs. The academy provides courses in Python and Robotics to primary and secondary students. The academy also provides classes in Python to adults and persons looking to upskill or gain a new skill in software development. The academy coding bootcamp program has graduated over 200 students since its inception and provided training in software languages to over 200 students on the island. Today, the academy has software developers working for foreign companies and individuals abroad. The academy has also participated in international coding hackathons and competitions abroad and three of its software applications were selected for the finals.

INCUBATION AND ACCELERATION PROGRAMMES

A further opportunity for dynamic cooperation comes through incubation and acceleration programs, which may be driven by either the public or private sectors.

ECOSYSTEM STAKEHOLDERS

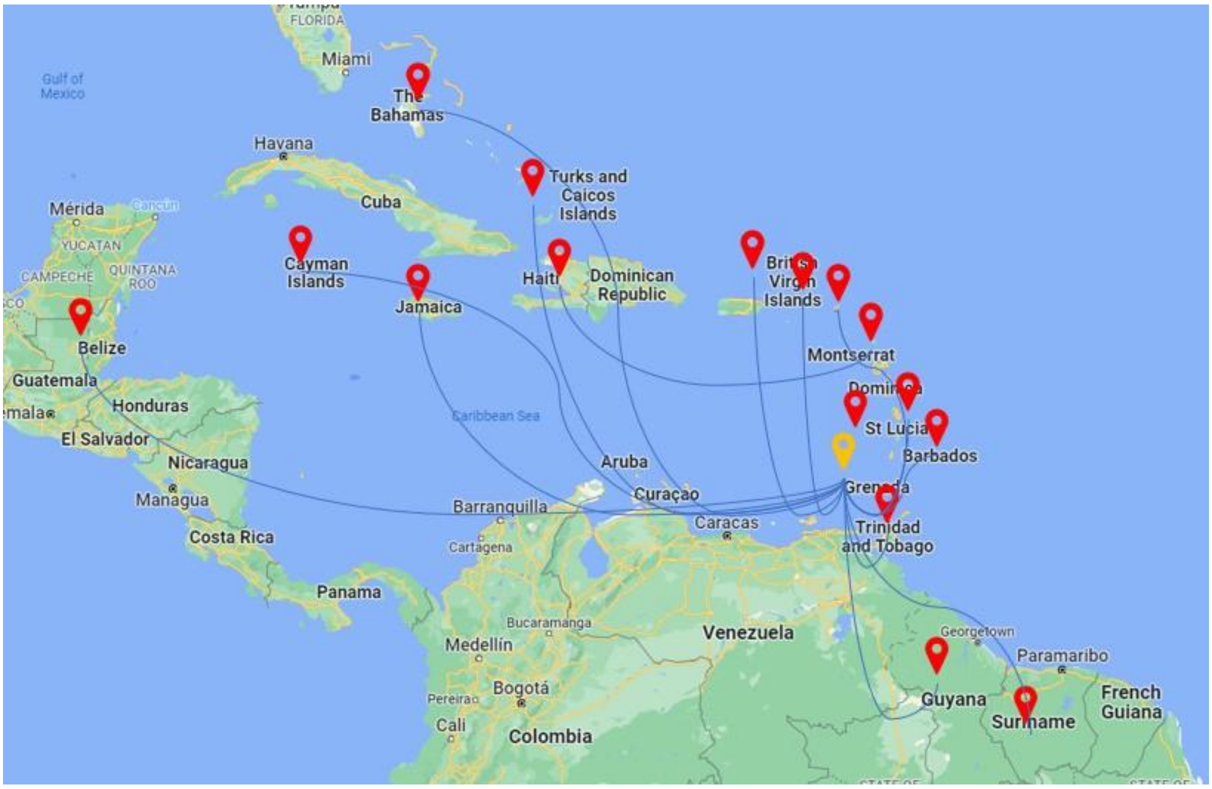
The success of the Technology & Innovation Hub relies on the creation of a self-sustaining ecosystem. During our research the team identified many stakeholders playing different roles at unique stages of the establishment of the ecosystem. Below, a non-exhaustive list of the roles and the value they will bring to the ecosystem.

ROLE	VALUE ADDED
GOVERNMENT	Policy makers and advocacy
EDUCATORS	Accredited and vocational training providers
INCUBATORS & ACCELERATORS	Local, regional and international programme providers
COMMERCIAL PARTNERS	Service providers i.e.: cloud storage, mentorship, advisors
FOUNDERS	Build solutions based on needs and advances in technology
INVESTORS	Angels, VCs, Institutional investors can provides access to capital to fuel innovation creation and growth
DIASPORA	Industry experts for mentorship, advisors
INTERNATIONAL ORGANIZATIONS	Technical advisory support, co-hosting of activities (training, challenges, etc.) global networks

CONNECTING THE INNOVATION & TECHNOLOGY HUB TO CARICOM

A final key proposed partnership is between Grenada and the rest of the CARICOM nations, where the aim is to connect a purpose-driven concept to others in the region.

The Technology & Innovation Hub will be a physical space that innovates the development of technological solutions to inherent social problems experienced by Grenada and other countries in the Caribbean. By using an open innovation and expandable mindset and idea strategy, other countries can adapt their own model Tech Hub through collaboration and partnership.



ROADBLOCKS TO INNOVATION

While Grenada is in an excellent position to move forward with developing a Technology & Innovation Hub, as with all multi-faceted plans, there are challenges to execution. Below is a summary of some of the projected initial challenges.

LACK OF AWARENESS OF OPPORTUNITIES WITHIN THE TECH SECTOR

As Grenada looks to increase the number of students becoming interested (and educated) in tech in all levels of their schooling, it is a challenge when students do not see tech as a viable career option in Grenada. Raising awareness and understanding of potential will be important as the Hub is built out.

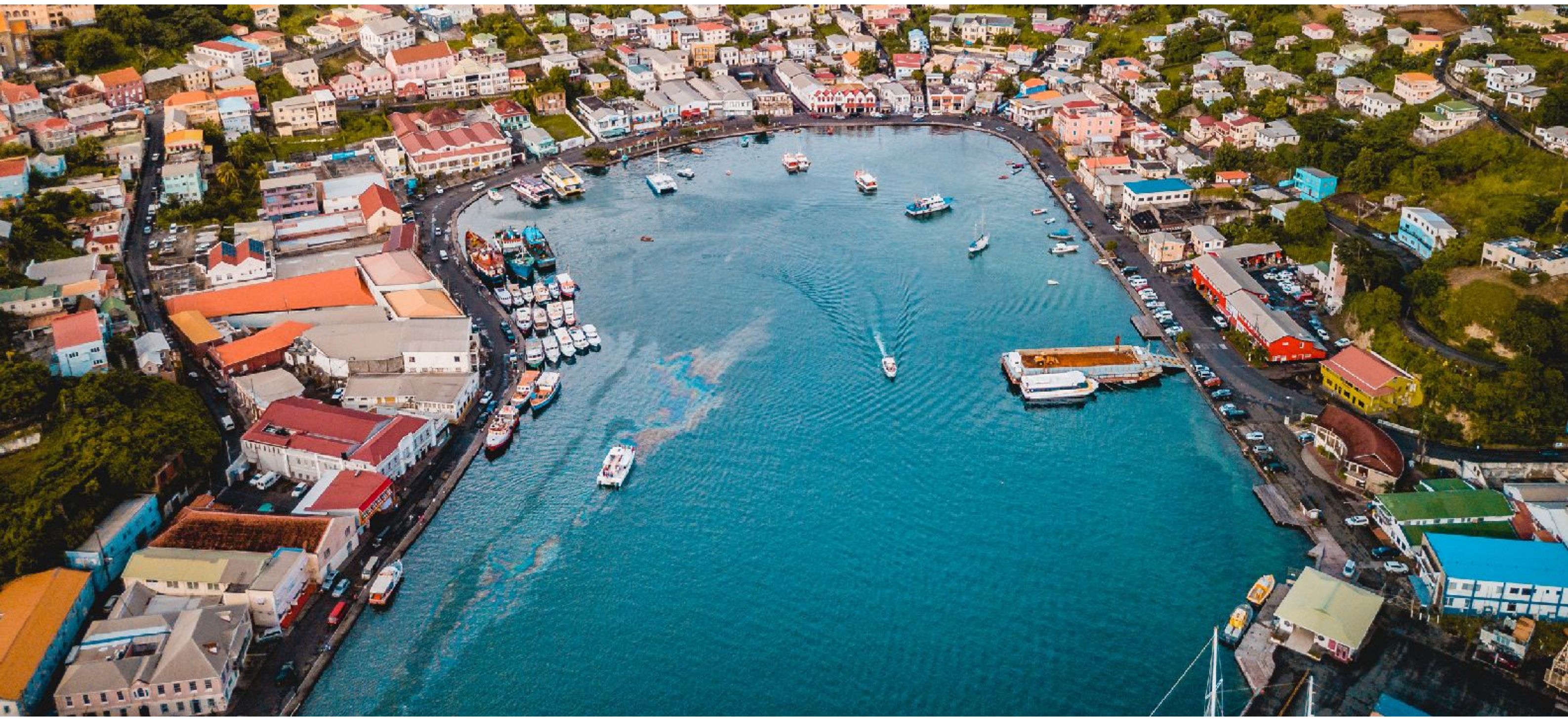
HIGH COST OF HARDWARE, SOFTWARE, AND TRAINING

Though a career in tech may be kicked off with fewer overhead expenses than a brick and mortar store or resort, restaurant, or similar, thanks to lower initial capital and inventory costs, the expense related to entering the tech industry must not be understated.

The current cost for high spec developer-build laptops and desktops averages to $2,000–$3,000 USD. This cost is certainly not negligible for those who are earning in Eastern Caribbean Dollars (XCD) and wishing to embark on a career in tech.

Additionally, there are relatively high ticket prices for training and certification courses such as SCRUM Master, Agile Coach, Project Management, Business Analysis, Product Management etc.

Without mitigating this particular challenge it will be a struggle to take full advantage of the talent and potential knowledge base living in Grenada.



FINANCING THE PROJECT

A project of this scope requires micro and macro level funding from a variety of sources. The team has identified several potential funders as well as outlining budget categories for the first several years and phases of the Technology & Innovation Hub's development.

An increase in investment in education by the government as well as effective investment incentives for foreign partners and investors building research and development (R&D) centres are paramount. Sustainable investment in technologies can positively impact Grenada's agriculture and tourism sectors as well as creating new fintech opportunities such as in crypto currencies and blockchain technology.

FINANCIAL ASSESSMENT

The team suggests the following options to finance the initiative:

- The World Bank "Caribbean Digital Transformation Project has allocated a total of USD$4.5 million dollars to digital skills and technology adoption from the USD$8.0 million allocated to Grenada.
- Diaspora and Community $200 Tech Innovation Development Bond, this can be achieved by targeting Grenadians to invest in this bond, potentially raising millions for the financing and setting up of the Technology & Innovation Hub.
- Citizen By Investment (CBI) program: investors under this program can be given the option to sponsor or contribute funds towards the Grenada Technology & Innovation Hub.
- Grant funding from wealthy organisations and companies like Google, Facebook and Microsoft etc.
- Line Ministry budgets could have allocations directed towards financing relevant activities of the Hub to offer impact in their respective sectors.
- Grenada Development Bank should play a pivotal role in managing the bond and other financial activities of the Hub. This will provide the necessary financial support or capital to enable a vibrant innovation and tech business community.

APPROACH TO FINANCIAL ASSESSMENT AND PROJECTIONS

This section provides information on the projected start-up and running costs of the Technology & Innovation Hub for its first three years as well as the anticipated revenues and contributions of founding partners over the same time period.

The revenue and expense data in this report is based on projections provided by the authors of this document, as well as market research of current prices found in Grenada for the goods and services outlined.

All calculations are in XCD dollars for consistency. The founding partners' contributions are based on extensive discussions within the team. The estimates are based on those discussions and are consistent with the type of information obtained from parties elsewhere.

PROJECTED EXPENSES AND REVENUES

A projected budget of $25-30 million dollars is recommended for this type of initiative based on our review of capital expenditures of several case studies throughout the world, however, we recommend that the development of this project can be scaled over a period of eight (8) years and this is based on availability of capital resources for a project of this type. As such, we are recommending an initial investment of $10,500,000 to start the project, please see projected and expenses breakdown.

The estimated seed capital needed to start the Technology & Innovation Hub is $10,500,000. From this amount, $4,500,000 will be allocated towards renovation and construction of the Centre. The full amount needed to sustain the project during its first three years of operation is $3,500,000. An estimated $500,000 is forecasted to come from revenues derived from fees for services to local and foreign companies, from partners' contribution and vendor fees totaling $1,500,000. The table below provides details of the total projected expenses for the first three years.

Table 1: Total Operating Revenue Forecast, Years 1–3

REVENUE SOURCES (TECH INNOVATION HUB	YEAR 1/ XCD$	YEAR 2/ XCD$	YEAR 3/ XCD$	TOTAL/ XCD$
TRAININGS— TECHNICAL SKILLS	$70,000	$140,000	$285,000	$495,000
TRAININGS—SOFT SKILLS	$145,000	$290,500	$580,000	$1,015,000
TRAININGS— SPECIALISED SKILLS	$160,000	$320,000	$640,000	$1,120,000
BUSINESS DEVELOPMENT SERVICES	$80,000	$160,000	$320,800	$321,040
CONSULTING AND PROOF OF CONCEPT	$200,000	$400,000	$800,000	$1,400,000
RENTAL — VENUE & EQUIPMENT	$25,000	$50,500	$100,250	$175,750
SPONSORSHIP FEES— GOLD	$40,000	$180,000	$360,000	$580,000
SPONSORSHIP FEES— SILVER	$80,000	$320,000	$640,000	$1,040,000

Table 2: Total Projected Expenses, Years 1–3

START—UP BUDGET ITEMS	XCD$
CONSTRUCTION OF NEW BUILDING	$4,500,000
EQUIPMENT (HARDWARE, SOFTWARE)	$1,000,000
OFFICE FURNITURE	$500,000
STAFF COST/PAYROLLS*	$1,500,000
UTILITY EXPENSES	$120,000
MARKETING	$60,000
TRAVELLING AND TRANSPORTATION	$40,000
EVENTS AND CATERING	$30,000

It is expected that the initial exploratory team members will continue to contribute to the Technology & Innovation Hub in a supportive and advisory role. Also, it is expected that the number of local companies including the public sector that will want to partner with the Hub will grow and that their collaboration will become increasingly significant. In addition, the Technology & Innovation Hub will proactively look for collaboration with donors on their development projects, and act as a third party in public projects and through other income-generating activities. The Technology & Innovation Hub may also generate revenue from services and products that have been developed by companies with Centre assistance.

Table 3: Example Income Statement

DESCRIPTION REVENUE	YEAR 1/ XCD$	YEAR 2/ XCD$	YEAR 3/ XCD$	TOTAL/ XCD$
FEE REVENUE	$150,000	$300,000	$600,000	$1,050,000
GRANTS	$500,000	$700,000	$950,000	$2,150,000
TOTAL REVENUE	$650,000	$1,000,000	$1,550,000	$3,200.000
EXPENSES	$144,000	$154,000	$170,000	$468,000
SALARIES	$600,000	$750,000	$800,000	$2,100,000
MARKETING	$36,000	$36,000	$36,000	$108,000
COMPETITIONS AND EVENTS	$10,000	$10,000	$10,000	$30,000
OFFICE SUPPLIES	$15,000	$18,000	$20,000	$53,000
DEPRECIATION	$10,000	$10,000	$10,000	$30,000
CATERING	$6,000	$6,000	$6,000	$18,000
TRAVEL—INTERNATIONAL	$20,000	$40,000	$60,000	$120,000
TRAVEL—DOMESTIC	$5,000	$8,000	$11,000	$24,000

PROPOSED INVOLVEMENT OF THE GOVERNMENT OF GRENADA

It has been the government's explicit goal to identify the optimal strategy and design plan of a Technology Innovation Hub that can be included into Grenada's Digital Transformation. Grenada is at a crucial point of opportunity for that strategy, and both the exploratory team, partners, and government will have important roles to play. This section identifies opportunities for the government of Grenada to support the Hub, perhaps through providing support and the policies to advance a technology and innovation agenda by executing thoughtful and efficient interventions. This will depend on establishing an appropriate and supportive regional partnership and infrastructure.

GOVERNMENT POLICY AND INITIATIVES FOR A TECHNOLOGY & INNOVATION HUB

Policy engagement can take a wide variety of forms. It could be advocating for a specific technical solution or policy change. It could be participating in the development or execution of policies. Or, it could be a more general shift in perspective and relationship between governmental and nongovernmental actors. Within that broad concept is the co-creation of a Technology & Innovation Hub.

Encouraging growth, increasing productivity, and promoting local technological innovation in the tech sector should be of the utmost importance to the Government of Grenada. To accomplish all these activities in today's fast-moving market requires agile and flexible government policies. The primary objective of the Technology & Innovation Hub is to encourage innovation for inclusive growth. This will eventually lead to increases in productivity, technological innovation and research and development (R&D).

The activities that the Hub could engage in are surrounded and shaped by the regulatory environment. This means that government policies and actions around innovation in general, or certain sectors like tourism, health and tech education more specifically, can either help or hinder an idea's progress.

For example, in Costa Rica the importation of computer hardware and software are tax exempt for everyone. This policy created many benefits and opportunities for locals and foreign companies operating or doing business in Costa Rica.

The Grenada Government can also look at the possibility of providing 8-10 years tax exemption and concession to companies transferring their businesses under **Grenada Tech Innovation Initiative.** The opportunities and challenges of direct engagement in policy co-creation are important to all stakeholders. As such, the team is suggesting ongoing collaboration with the Technology & Innovation Hub to achieve policy changes in the following areas:

- **Increase investment in education and job training,** specifically in the field of Science, Technology, Engineer and Mathematics. Chancellor Charles Modica from St. George's University and Dr. Keith Mitchell spoke about the importance of getting persons trained in the field of Information Communication and technology (ICT) and the new digital economy. Government needs to take the leadership role in creating a professional community of computer science lecturers or teachers. Introducing computer science education (Coding and Robotics) into the primary and secondary schools. In short, we need to prepare our students for life in this digital age.

- **Provide incentives or changes in immigration policy** to attract (key experts and non-key experts) the necessary human capital and for the transfer of knowledge.

- **Provide Targeted and Effective Investment Incentives** to foreign and local investors to start new businesses and create services and products that can be exported.

- **Take the lead in providing Investment in Research and Development (R&D)** in agriculture, the blue economy and other key economic sectors. Define a new development strategy, building on the Small Smart State document by creating a highly skilled labor force, and a culture of technological and scientific excellence.

- **Solicit International Support for R&D programs,** which are joint research and development programs with foreign partners, for example, UNDP, USAID, and UNESCO. These joint R&D programs can be in the areas of agriculture, tourism, transportation and alternative medicine.

- **Support for an Incubator and venture capital programs** supported by Grenada Development Bank (GDB) to convert promising research and innovation into successful businesses.

- **Finance Innovation,** heavy investment into technology incubators in light manufacturing, assembling and product and service development.

Research shows that a Technology and Innovation Hub can have positive implications for job creation and new systems and services, many of which are ICT-enabled. Government thus has the opportunity to create additional employment through a Technology and Innovation Hub ecosystem and to ensure that they are able to attract and sustain investments and participation in the ICT sector, while simultaneously ensuring the availability of a local talent pool.

Policies should strike a balance between encouraging basic research, applied research, and innovation models. For the Grenada Technology & Innovation Hub, investment in research and innovation is necessary towards supporting applied research and commercialization projects.

Designing policy measures to support entrepreneurial commercialization of science and technology is important for this Technology & Innovation Hub; policymakers need to acknowledge that exponential growth would not be the only form of business development that leads to new jobs and wealth. Different forms of growth have different limitations. For instance, when growth happens through acquisition and rapid expansion, access to risk finance can be a bottleneck for further development.

The availability of venture capital funding is crucial to allow companies to grow and innovate, especially for start-ups and innovative small, medium enterprises (SMEs). In more general terms, consistent with the presence of different phases of growth described in this study, government policies can play a role in “bridging” industries (within high-tech sectors or between high-tech and traditional sectors). Policymakers can use their influences to encourage cross-sectorial innovation with neighbouring islands, and at the same time to encourage and share the cost of tech-enabled innovation.

One example of a bridge between technology and industry can be seen in agriculture. Using automation and robotics in the agriculture sector may have great immediate impacts and benefits. Robots may be used for weed prevention, cloud seeding, environmental monitoring and soil analysis. Some examples of weed prevention are robots that only target weeds using vision technology. The accuracy of the technology and capabilities of running on solar energy can make a sustainable solution reducing the amount of labor for eliminating weeds and other plants that may put crops in danger.

Some examples of environmental monitoring may be seen in drone technology. Drones are well equipped with vision technology that allows the user to see a vast amount of the crop environment to provide accurate data and feedback to farmers. Using real-time data, farmers are able to diagnose crops without setting foot on site. This data may be used to prevent further loss to crops from animals, disease, and other contaminants.

Policymakers can create the conditions for the growth and the diffusion of a strong Tech Open Innovation culture. The Technology & Innovation Hub will organise initiatives to prepare engineers, doctoral students, and managers for the implementation of open innovation strategies such as alliances and technological co-development. This sharing enables the generation of wealth based on knowledge, skills, open competition, increased capacity, and efficiency.

Government policies can help to release astonishing development of a Grenada Technology & Innovation Hub as it did with Israel's Tech sector as well as the whole of its high-tech industry. The key to innovation is the capability to establish and cultivate a culture in which young people are not afraid to establish their own startups. This requires capable human capital as well as a supportive business environment. A digital environment where the government, individuals, organisations, and communities are empowered by the availability of information, access to it, and the means to share, analyse, and to generate knowledge through such interaction is the goal.

Potential Government support for the creation of a Physical Space

The team has identified several possible locations in St. George's for hosting the Hub locations; this information was provided by the Minister of Parliament (MP) Mr. Peter David. The location provided was Grenada York House on Church Street and the Public Library on the Carenage. For instance, the Government of Grenada could provide the building on a lease contract for this type of project, with the not-for-profit organization responsible for renovations.

The buildings identified need intensive renovation and resources for them to be functional or operational. The other option for a physical space would be to construct a new building on land that can be provided by the Government. A virtual space can be implemented very quickly, however, it is important that Grenada couples that virtual space with a physical office space. Grenada Olympic Building was also identified for the physical space or office.

Political Continuity

While Grenada is politically very stable, partisan party politics can significantly derail any progress made should the ruling party change during the course of the implementation of the Hub if sufficient policy protections are not put in place to ensure continuity through transition.

Finally, the government can contribute significantly by working to smooth or eliminate some specific pain points for both established and new and emerging businesses, identified below:

Business Registration

The process of registering a business within Grenada involves the completion of multiple processes from several disjointed agencies. Coupled with a governmental website of unpleasant layouts and a broken search function makes the process far more difficult when compared to the processes for other countries.

Following the issuance of the certificate of name registration, your business must now be registered for a taxation ID number. Due to the disjointed nature of the agencies and processes, it is now necessary to register the Tax ID number with the Inland Revenue office. Thankfully this process can be done online through the government website, although that is easier said than done. The convoluted nature of this process is further exacerbated by the inefficiency of the government website; any attempt to use their search function results in an error message screen and the subsequent agency's steps are not available through the previous section's page. This makes it extremely tedious to understand the correct processes involved in registering the business and due to the disjointedness of the agencies, each one must be contacted separately for their specific process information.

Once that registration has been successfully tackled, businesses must turn to the National Insurance Scheme (NIS). Fortunately, and as with the previous section, this can be conducted through the online self service portal. Once the certification has been submitted to the NIS organisation, the business will be successfully registered with NIS and the process is now complete.

Banking

While banks in Grenada do often offer an eService banking option, none offer an option to set up bank accounts online. As businesses continue to look into external markets or eCommerce websites, the need for inexpensive online payment gateway effective digital tools for financial processes is essential.

As if to mirror the initial process of name registration; the presentation of documents such as the source of funds declaration must also be done in person. The banks' inability to provide this basic service through digital tools is a clear indicator of their lack of technological capacity and a sure deterrent of future development. The process of applying for a loan in Grenada is likewise limited by the need to visit a physical branch. In many cases, the information provided on the website is limited and not very clear. For example, the benefits and general information of a Term Loan is listed on the Republic bank website while the actual process is absent.

While the team recognizes that government and banking industries stand alone, government support to develop and streamline the process will benefit all Grenadians, regardless of their involvement with tech.

TEAM AND ORGANISATIONAL STRUCTURE

The report team of local and international experts propose to support the launch and coordination of the Technology & Innovation Hub as and where possible. They envision the launch to be supported by industry, education and government institutions. This support may include:

- Coordinated training between St. George's University, Caribbean Coding Academy, and the University of the West Indies Open Campus, as well as other institutions locally

- Coordinating the global marketing of the Grenada Technology & Innovation Hub with the Ministry of Tourism and others

- Tapping into the expertise of Grenadians now living abroad and others with connections to the islands through the "Global Grenadian" community. Modelled after the GlobalScot program it can be used as a "reverse brain drain" program encouraging expats to connect back with the local community, wherever they may now live.

WHO ARE GLOBAL SCOTS

According to the GlobalScot website, "GlobalScots are passionate individuals, entrepreneurs and business leaders, who are dedicated to supporting Scotland's ambitions internationally.

They are:

- Based anywhere in the world but have a connection to Scotland.
- Willing to share their knowledge, insights and networks with ambitious Scottish businesses and organisations.
- Inspirational individuals who can make a significant difference to Scotland's global competitiveness."

A similar network of worldwide Grenadians and those connected to the country would provide a supportive springboard for new and burgeoning projects and enterprises emerging from Grenada.

MANAGEMENT

Competitiveness in today's high-paced world does not depend solely on the economic factors of production, technological advantage, or on a specific resource. Rather, it depends on a supportive and enabling environment, high-level skills and learning, and continuous innovation.

An Innovation Steering Committee of founders is a powerful coalition of leaders that are key experts and professional individuals with a broad range of experience who are capable of guiding and overseeing the entire journey of innovation for this endeavor. They would be responsible for driving the processes of change that will move this organisation towards a culture that supports and sustains innovation; constantly monitoring it to make sure that it stays on track. The Innovation Steering Committee founders would be a permanent body; its members should include a mix of executive management. Following is an explanation of what it means to be a committee member; the roles and involvement:

Committee founder members must share the commitment, responsibilities, mission, vision and equity of the organisation. The founders would be required to sign a memorandum of understanding (MOU), with regards to responsibility, policies, duties, liability and risk exposure. After signing this agreement, committee founders will proceed to incorporate a non-for-profit legal entity.

Roles: Chief Executive Officer (CEO), Chief Technology Officer (CTO), Chief Operations Officer (COO), Chief Marketing Officer (CMO)

Management activities: Business plan development, establishing a mission and vision, becoming an innovation architect, forming the board of directors and executive team, maintaining a relationship with the board of directors, overseeing the daily operations of the organisation, providing initial funding, and being public spokespersons for Grenada's Technology & Innovation Hub.

As an interim step, an exploratory team has been formed to undertake the research and writing of this report, and shepherd some logistical and exploratory steps predating the formation of a core team.

MEMBERS OF THE EXPLORATORY TEAM

CLIFFORD BROWNE

- Founder and CEO of Caribbean Coding Academy
- Bachelor of Science in Finance and Accounting
- Master's degree in Energy management

ALEXANDER (SASCHA) WILLIAMS

- Executive leader with extensive experience in high-growth tech start-ups, tech community building, strategic project development and global corporate management
- Helps tech company entrepreneurs with their growth strategies as board and executive advisor
- Formerly: Electronic Arts, Microsoft (BigPark), the Walt Disney Company
- Helped establish a similar Tech Hub in a fast-growing tech centre in Canada (see case study on page 17)

GISELLE FREDERICK

- Analytical, multidisciplinary technologist with extensive cross-functional experience consulting with large firms, managing change projects and leading start-ups.
- Represents the UK at forums such as G20 YEA, UN Women and mentors at start-up Accelerators in the UK and across Europe: Foundervine, Cornerstone 2.0, Hatch CoLab, Google for Startups, and Huge Thing.
- Founder of three mission-driven start-ups (Sonaaar, CoVolEx, Zingr)

Dr. CARL S. ROBERTS

- Over 20 years of experience working with billion-dollar organisations in the financial and healthcare industries
- Senior Analyst/Senior IT Infrastructure Engineer.
- BS—Bachelor of Science in Information Technology: General
- MSIT—Master of Science in Information Technology, Specialization: Strategic Leadership in IT
- DIT—Doctor of Information Technology | Specialization: Tech | Cloud/Grid Computing
- Foundational member of the GBSS Alumni International Foundation Inc. (GBSS-AIF); A 501c3 non-profit, charitable organisation.

FILIP JAKUBOWSKI-DRZEWIECKI

- Previously the CTO of one of the most vibrant UK branchless neo-banks
- Currently supports new initiatives helping SME to manage and plan their finances (Helu) as well as impact investing (The Gather App).
- Contributes to the community by supporting start-ups via accelerators like Startupbootcamp (Fintech & Insuretech) and Google Launchpad.

RICHI R. RAMLAL

- Transformational leader in the spheres of innovation, technology and culture.
- Bachelor's of Science in Mechatronics Engineering.
- Experienced in new and emerging technology, he leads technological advancements for manufacturing companies.
- Began his career in the United States Military as a helicopter flight engineer, where he discovered his love for innovation and technology.

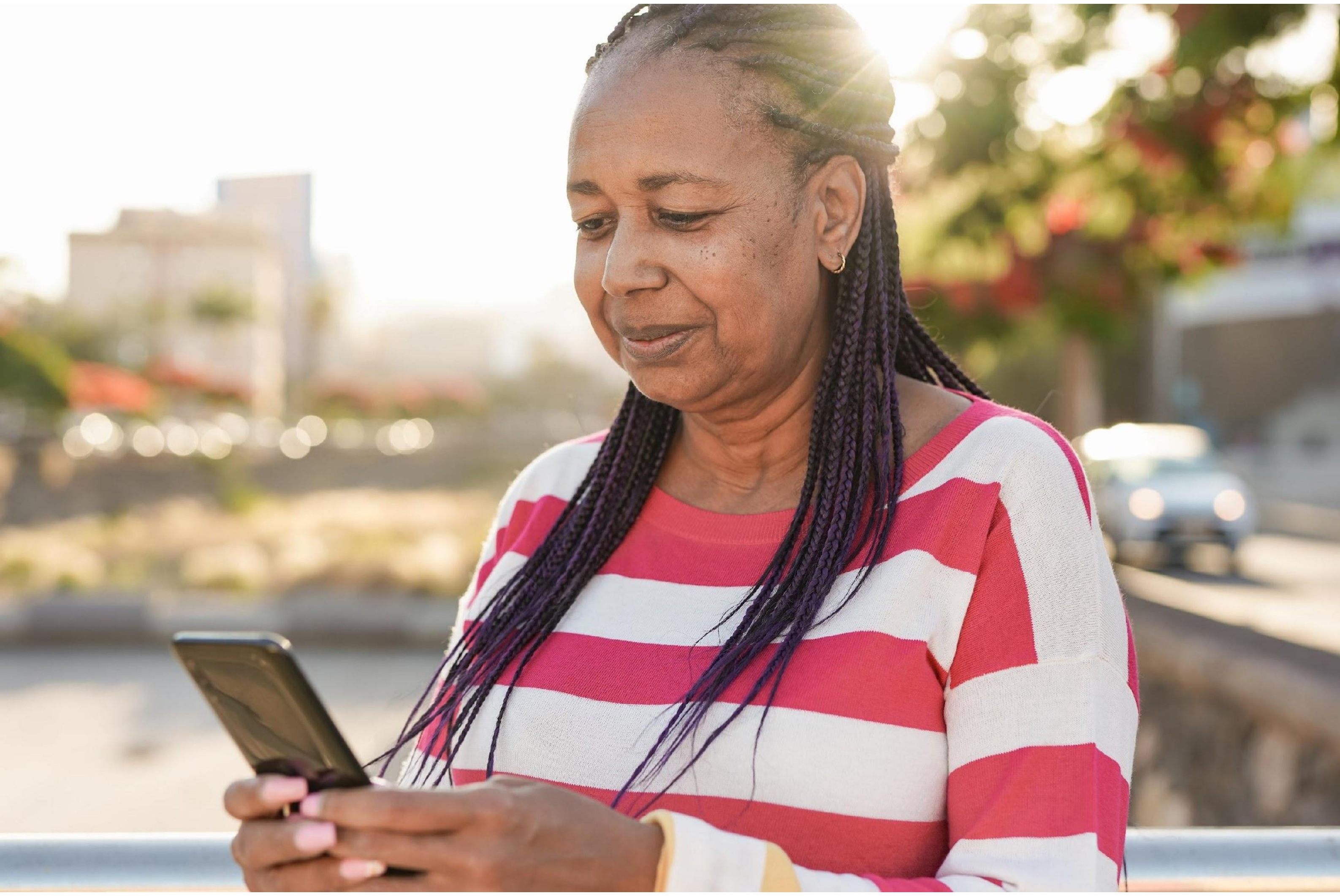
GEORGE JR MATTHEW

- Has a wealth of experience as an academic, innovator, researcher and regulatory executive.
- Has 17 years of extensive practical, managerial, research, and technical analysis experiences within sustainable island electricity systems modelling, investments, policy, regulation and innovative technology.
- BSc (Magna cum laude) in Mathematics and Physics
- Post-Graduate studies in Wireless Communications
- Doctorate in Sustainable Low-Carbon Electricity Systems Policy, Regulation and Investment.
- Currently acts as a consultant energy advisor to support the energy work program of the World Bank in the Caribbean region
- Director of the Caribbean Coding Academy

CONCLUSION

The path towards a dynamic and visionary hub in the Caribbean, grounded in Grenada, is certainly not without challenge. However, the report team is confident that through driven teamwork, internal and external expertise, and government engagement, the development of a Technology & Innovation Hub in Grenada would lead to something truly groundbreaking in the region.

Grenada has the opportunity to up skill a nation while offering unique and much-needed knowledge-based capital to the world. The exploratory team has laid the groundwork and identified possible next steps to bring a Technology & Innovation Hub to Grenada. Are you ready to come on board?



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